

30 Industry-Relevant Quant Finance Projects

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If you're targeting quant research, market risk, model validation, derivatives pricing, or credit roles, here are 30 industry-relevant projects ranging from intermediate to advanced.

Market Risk

1. Historical Simulation VaR & Expected Shortfall
2. Parametric VaR with EWMA and GARCH Volatility
3. Monte Carlo VaR using Full Portfolio Revaluation
4. Stress Testing and Scenario Analysis (2008, COVID-19, Inflation Shock)
5. Backtesting VaR (Kupiec, Christoffersen, Basel Traffic Light)

Derivatives Pricing

6. European Option Pricing using Black-Scholes
7. American Option Pricing using Binomial Tree (CRR & Leisen-Reimer)
8. Monte Carlo Pricing of Asian, Barrier and Lookback Options
9. Finite Difference Method (Explicit, Implicit & Crank-Nicolson)
10. Implied Volatility Surface Construction and SVI Calibration

Volatility Modeling

11. GARCH vs EGARCH vs GJR-GARCH Volatility Forecasting
12. Heston Stochastic Volatility Model Calibration
13. Local Volatility Surface using Dupire Model
14. SABR Model Calibration for Interest Rate Options
15. Realized Volatility Forecasting using High-Frequency Data

Fixed Income

16. Yield Curve Construction (SOFR/OIS & Treasury)
17. Bond Pricing, Duration, Convexity and Key Rate Duration
18. Interest Rate Swap Pricing and Risk Analysis
19. Hull-White and Vasicek Interest Rate Models
20. Mortgage Prepayment and PPNR Time Series Modeling

Credit Risk

21. Probability of Default (PD) Model using Logistic Regression and XGBoost

22. IFRS 9 Expected Credit Loss ($PD \times LGD \times EAD$)
23. LGD Modeling using Jacob Frye Method
24. Credit Portfolio VaR using CreditMetrics
25. Survival Analysis (Kaplan-Meier & Cox Model) for Loan Defaults

Quantitative Research

26. Statistical Arbitrage (Pairs Trading using Cointegration)
27. Factor Investing Model (Value, Momentum, Quality, Low Volatility)
28. Machine Learning Return Prediction (XGBoost, LightGBM, LSTM)
29. Time Series Forecasting using ARIMA, SARIMA, VAR and LSTM
30. Reinforcement Learning Trading Agent (DQN/PPO) with Transaction Costs

Bonus Advanced Projects (used by many hedge funds and banks)

- XVA (CVA, DVA, FVA, MVA) Monte Carlo Engine
- STANS-style Portfolio Margin Simulation
- Copula-based Portfolio Risk Modeling
- Dynamic Delta-Gamma-Vega Hedging System
- Multi-Asset Risk Engine for Equities, Options and Bonds
- Exotic Option Pricing using Longstaff-Schwartz Least Squares Monte Carlo
- Portfolio Optimization using Black-Litterman
- Market Regime Detection using Hidden Markov Models
- Bayesian Portfolio Optimization
- Limit Order Book Modeling and Market Microstructure Analysis

PS: If you want to stand out, build projects that demonstrate depth rather than just familiarity.

Source: [LinkedIn post](#) by Mehul Mehta (Lead Quant at OCC).